

Thesis Proposal

SOP-Guard

*An Agentic RAG System with Procedural Faithfulness Verification
for Clinical Standard Operating Procedures*

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1. Introduction

Hospitals operate on the basis of Standard Operating Procedures (SOPs) — detailed, step-by-step documents that govern clinical workflows ranging from sepsis management and blood transfusion protocols to medication administration and infection control. These documents are the backbone of patient safety and clinical quality assurance. A typical tertiary hospital maintains hundreds of SOPs across dozens of departments, each subject to periodic revision as medical knowledge evolves.

Despite their critical importance, accessing information within SOPs remains a manual, time-consuming process. Clinicians — particularly during emergencies, night shifts, or high-acuity situations — must search through lengthy PDF documents to locate a specific dosage threshold, verify a step sequence, or check a contraindication. This information retrieval bottleneck introduces delays in clinical decision-making and creates opportunities for error.

Recent advances in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have demonstrated the potential to transform how humans interact with document repositories. However, deploying such systems in clinical settings presents unique challenges. A standard RAG system can retrieve relevant passages and generate fluent responses, but it cannot guarantee that its output is **procedurally faithful** — that dosage values are accurate, that steps are presented in the correct sequence, and that critical contraindications are not omitted. In healthcare, such errors are not merely inconvenient; they are potentially life-threatening.

This thesis proposes **SOP-Guard**, an agentic RAG system specifically designed for clinical SOP question-answering that incorporates a novel **Procedural Faithfulness Verification** layer. The system employs multiple specialized AI agents orchestrated through a deterministic graph-based pipeline, supports voice input for hands-free clinical use, enables clinician-driven SOP updates with full version tracking, and provides transparent reasoning traces for every response. The goal is to build a system that is not only academically rigorous but practically deployable in real hospital environments.

2. Problem Statement

Current approaches to clinical SOP access suffer from three fundamental limitations:

- **Inefficient Information Retrieval:** Clinicians spend significant time manually searching through PDF-based SOP documents. Studies indicate that healthcare workers spend up to 25% of their time on information retrieval tasks, directly reducing time available for patient care.
- **No Procedural Verification in AI Systems:** Existing clinical QA systems and general-purpose LLMs can generate answers from medical text, but they lack mechanisms to verify that the generated answer is procedurally correct. An LLM might confidently state a dosage that is slightly off, present steps out of order, or omit a critical contraindication — errors that are difficult for busy clinicians to catch and potentially dangerous for patients.
- **Static, Disconnected Knowledge Bases:** Hospital SOPs are living documents that should evolve with clinical experience and new evidence. However, the feedback loop between frontline clinical knowledge and SOP documentation is slow and fragmented. Clinicians who discover better practices or identify outdated protocols have no streamlined way to contribute updates back to the knowledge base.

This thesis addresses all three limitations through a unified system that combines intelligent retrieval, automated procedural verification, and a structured clinician feedback and update mechanism.

3. Research Questions

This research seeks to answer the following questions:

***RQ1:** How effectively can a Procedural Faithfulness Verifier detect threshold errors, sequence violations, and contraindication omissions in LLM-generated clinical SOP responses?*

***RQ2:** Does an agentic, multi-agent architecture with verification feedback loops produce more accurate and safer clinical SOP answers compared to standard single-pass RAG systems?*

***RQ3:** How does clinician feedback integration and structured SOP version management impact the quality and currency of the system's knowledge base over time?*

***RQ4:** What is the effect of voice-based input on the usability and adoption of clinical decision support systems in real hospital workflow conditions?*

4. Research Objectives

O1. Design and implement SOP-Guard, an agentic RAG system with procedural faithfulness verification for clinical SOP question-answering.

O2. Develop a structured SOP representation schema that enables automated verification of thresholds, step sequences, and contraindications.

O3. Build and evaluate a Procedural Faithfulness Verifier capable of detecting three categories of clinical errors: threshold violations, sequence violations, and contraindication omissions.

O4. Implement a voice input interface using speech-to-text technology optimized for medical terminology, enabling hands-free clinical queries.

O5. Design a clinician-driven SOP update portal with version tracking, approval workflows, and automatic knowledge base re-indexing.

O6. Create an adversarial evaluation dataset of procedurally incorrect clinical responses for benchmarking verification systems.

O7. Evaluate the complete system against baseline RAG approaches using both automated metrics and expert clinical review.

5. Literature Review

5.1 LLMs in Clinical Decision Support

Large Language Models have shown remarkable capability in medical question-answering, with models like Med-PaLM 2 achieving expert-level performance on USMLE-style questions (Singhal et al., 2023). GPT-4 and Claude have demonstrated strong clinical reasoning across a range of medical benchmarks. However, these models are general-purpose and lack domain-specific safeguards for procedural accuracy. Studies have shown that even high-performing medical LLMs can hallucinate dosages, invent contraindications, or present clinical steps in incorrect order (Ji et al., 2023). The gap between fluent generation and procedurally safe generation remains a significant barrier to clinical deployment.

5.2 Retrieval-Augmented Generation (RAG)

RAG systems address the hallucination problem by grounding LLM responses in retrieved documents (Lewis et al., 2020). In clinical settings, RAG has been applied to medical literature QA and drug information retrieval. Hybrid retrieval approaches combining dense embeddings with sparse methods (BM25) have shown improved performance for technical medical queries where exact term matching (drug names, dosages) is critical (Ma et al., 2023). However, standard RAG provides no mechanism to verify that the generated response correctly reflects the retrieved evidence — the model can still misinterpret, selectively omit, or incorrectly combine information from source passages.

5.3 Agentic AI Systems

Recent work on agentic AI architectures has demonstrated the value of decomposing complex tasks into specialized sub-agents. Frameworks like LangGraph (LangChain, 2024) enable deterministic, graph-based orchestration of multiple LLM agents with conditional routing, retry loops, and state management. This approach is particularly valuable for safety-critical applications where predictable control flow and auditability are essential. Unlike autonomous multi-agent frameworks (AutoGen, CrewAI), graph-based orchestration provides the determinism required for clinical systems.

5.4 Faithfulness and Hallucination Detection

Faithfulness evaluation in NLG systems has been studied extensively (Maynez et al., 2020). Metrics like RAGAS (Es et al., 2023) provide automated faithfulness scoring for RAG systems. However, existing approaches evaluate faithfulness at the semantic level — whether the answer is generally consistent with the source. They do not evaluate **procedural faithfulness**: whether specific numerical thresholds are preserved, whether step sequences are maintained, and whether conditional logic (contraindications, exceptions) is correctly represented. This gap is precisely what our Procedural Faithfulness Verifier addresses.

5.5 Voice Interfaces in Healthcare

Voice-based clinical tools have gained attention with the maturation of speech-to-text systems. OpenAI's Whisper model achieves near-human accuracy in general transcription and can be fine-tuned for medical vocabulary (Radford et al., 2023). Voice interfaces offer significant ergonomic benefits in clinical settings where hands are occupied. However, challenges remain: hospital environments are noisy, medical terminology has high phonetic similarity between critical terms (e.g.,

"heparin" vs. "Hespan"), and privacy concerns exist when queries are spoken aloud near patients. Our design mitigates these risks through transcription confirmation before processing.

5.6 Research Gap

To our knowledge, no existing system combines (a) RAG-based clinical SOP question-answering with (b) automated procedural faithfulness verification, (c) agentic multi-agent orchestration with verification feedback loops, (d) voice input for hands-free use, and (e) a clinician-driven SOP update mechanism with version tracking. SOP-Guard addresses this gap as an integrated system.

6. Proposed System — SOP-Guard

6.1 System Overview

SOP-Guard is an AI-powered clinical decision support system that enables healthcare professionals to query hospital SOPs using natural language (typed or spoken) and receive verified, sourced, and confidence-scored answers. The system is distinguished by its **verify-then-respond** architecture: every generated answer passes through a Procedural Faithfulness Verifier before being presented to the clinician. If the verifier detects errors, the system either corrects itself or escalates transparently.

6.2 System Architecture

The system processes each query through six sequential stages, orchestrated by a LangGraph state machine with conditional routing:

- **Stage 1 — Query Intake:** The clinician submits a query via text or voice. Voice input is transcribed using Whisper and displayed for confirmation. The Query Understanding Agent classifies the query type (threshold check, sequence question, contraindication lookup, general) and extracts clinical entities (drug names, conditions, parameters).
- **Stage 2 — Hybrid Retrieval:** The Retrieval Agent performs combined dense retrieval (embedding similarity) and sparse retrieval (BM25) over the structured SOP knowledge base, with metadata filtering by department and SOP category. A cross-encoder reranker selects the top-k most relevant passages. If retrieval confidence is below threshold, the agent reformulates the query and retries once.
- **Stage 3 — Answer Generation:** The Reasoning Agent synthesizes an answer strictly grounded in retrieved passages. Structured prompting enforces citation of source sections. For multi-step procedures, the output follows a numbered step format with decision branch points.
- **Stage 4 — Procedural Verification:** The Verifier Agent checks the answer against the structured SOP representation for three violation types: (a) threshold errors — are all numerical values correct? (b) sequence errors — are steps in the right order with none omitted? (c) contraindication errors — are stated exceptions properly accounted for? On failure, the answer is regenerated with the violation flagged as a constraint (maximum one retry, then escalation).
- **Stage 5 — Confidence Scoring:** A composite confidence score is computed from retrieval relevance scores, verifier results, and answer-source textual overlap. Three output tiers: high confidence (full answer), medium confidence (answer with advisory + SOP link), low confidence (answer withheld, clinician directed to source document or human expert).
- **Stage 6 — Response & Feedback:** The verified answer is presented with highlighted source passages, a step-by-step reasoning trace, and the confidence score. Clinicians can provide feedback (thumbs up/down + free-text correction) which feeds into the continuous learning loop.

LangGraph Orchestration Flow:

```
QueryIntake → Retrieval → [sufficient?] → Reasoning → Verification → [pass?] → ConfidenceGate → Output → Feedback
```

Conditional edges allow the verifier to route back to retrieval or reasoning, and the confidence gate to route to escalation — enabling self-correcting behavior within an auditable control flow.

6.3 Procedural Faithfulness Verifier (Core Contribution)

The Procedural Faithfulness Verifier (PFV) is the central research contribution of this thesis. Unlike general faithfulness metrics that assess semantic consistency, the PFV performs structured verification against a formal SOP representation.

The PFV checks three violation categories:

- **Threshold Violations:** All numerical values in the answer (dosages, time windows, lab ranges, vital sign thresholds) are extracted and compared against the structured SOP. Example: the SOP states "administer heparin 80 units/kg" but the answer states "60 units/kg" — this is flagged as a threshold violation with the specific discrepancy cited.
- **Sequence Violations:** For multi-step procedures, the PFV verifies that (a) all mandatory steps are present, (b) steps are in the correct order, and (c) no steps are incorrectly merged or split. Example: the SOP requires blood cultures before antibiotics, but the answer reverses this order.
- **Contraindication Violations:** The PFV checks whether the answer accounts for all stated contraindications and conditional exceptions. Example: the SOP states "do not administer metformin in patients with eGFR < 30" but the answer provides a general dosage without this caveat.

To enable this verification, each SOP is parsed into a semi-structured JSON schema containing ordered steps (with preconditions and time constraints), numerical thresholds (with parameter, operator, value, and unit), and contraindication rules. This structured representation is created through a combination of PDF parsing (PyMuPDF) and LLM-assisted structuring, with human expert validation.

6.4 Voice Input Interface

SOP-Guard supports audio-based queries for hands-free use during clinical procedures. Clinicians tap a microphone button on the interface, speak their question, and the audio is transcribed using OpenAI Whisper, fine-tuned on medical terminology. The transcribed text is displayed for confirmation before submission — this confirmation step is critical for safety, as it allows clinicians to catch transcription errors in drug names or dosage terms before the query is processed. The voice interface is designed for non-bedside use (nursing stations, break rooms) to mitigate patient privacy concerns.

6.5 Clinician-Driven SOP Update Portal

A key practical innovation of SOP-Guard is its structured mechanism for clinicians to contribute updates back to the SOP knowledge base. The system includes an update portal where authorized users (department heads, quality officers, senior clinicians) can:

- Propose modifications to specific SOP sections with a reason for the change.
- Each update is tagged with the clinician's identity, department, date, and a free-text justification.
- Proposed updates enter an approval workflow — reviewed by designated approvers before going live.
- The system maintains a complete version history, allowing anyone to see what changed, when, and why.
- Upon approval, the AI knowledge base (vector store and structured SOP representation) is automatically re-indexed, ensuring that subsequent queries reflect the latest protocols.

This creates a **living knowledge base** that evolves with clinical practice rather than becoming stale between periodic manual reviews.

6.6 Feedback & Continuous Learning Loop

Every response includes feedback controls. Clinician feedback is structured and logged with the query, response, clinician role, department, and correction text. This data serves three purposes:

- **SOP quality signal:** Recurring corrections on the same SOP section flag it for review.
- **Retrieval improvement:** Feedback is used to fine-tune retrieval ranking, adding hard-negative examples and adjusting relevance weights.
- **Institutional analytics:** Monthly digests show which SOPs cause the most confusion, which departments query most frequently, and where the system's accuracy is weakest — providing actionable intelligence for hospital quality improvement teams.

6.7 Explainability & Trust Mechanisms

Clinical adoption requires transparency. SOP-Guard provides multiple layers of explainability:

- **Source highlighting:** Every claim in the answer is linked to the specific SOP passage it came from.
- **Reasoning trace:** A step-by-step breakdown showing: which passages were retrieved → how the answer was composed → what the verifier checked and the result.
- **Confidence score:** A calibrated numerical score indicating the system's certainty, with clear communication when confidence is low.
- **Counterfactual explanations:** Where applicable, the system explains how the answer would differ under alternative conditions (e.g., "If the patient had hepatic impairment, the dose would be reduced to...").
- **Explicit abstention:** The system is designed to say "I am not confident enough to answer this — please consult the source SOP" rather than generate a potentially unsafe response.

7. Technology Stack

Component	Technology	Rationale
LLM (Primary)	Claude Sonnet 4.6	Strong reasoning, structured output support
LLM (Local/Fallback)	Llama 3.1 70B	On-premise deployment, no data leaves network
Voice Transcription	OpenAI Whisper	High accuracy, fine-tunable for medical terms
Agent Orchestration	LangGraph	Deterministic graph, conditional routing, auditable
Embeddings	text-embedding-3-large / BGE-M3	Semantic retrieval; BGE-M3 for on-prem
Vector Database	Qdrant	Hybrid search, metadata filtering, on-premise capable
Sparse Retrieval	BM25 (Elasticsearch)	Exact term matching for drug names, dosages
Reranker	ms-marco-MiniLM-L12	Cross-encoder reranking for precision
SOP Parsing	PyMuPDF + LLM structuring	PDF to structured JSON with expert validation
Backend API	FastAPI (Python)	Async support, OpenAPI docs, production-ready
Frontend	React + Tailwind CSS	Mobile-responsive PWA for clinical use
Database	PostgreSQL	Feedback logs, SOP versions, user management
Deployment	Docker + Kubernetes	Hospital data center deployment
Evaluation	RAGAS + custom metrics	Faithfulness, retrieval quality, safety scoring

8. Datasets & Evaluation Plan

8.1 Public Datasets

Dataset	Description	Use in This Work
PubMedQA	1,000 expert-labeled clinical QA pairs	QA evaluation benchmark
MedQA (USMLE)	12,723 clinical reasoning MCQs	Reasoning capability evaluation
MedMCQA	194K medical MCQs	Large-scale clinical knowledge testing
MIMIC-IV Notes	Discharge summaries (credentialed access)	Entity extraction validation
ExpertQA	2,177 expert-validated QA pairs	Answer quality benchmarking
MTSamples	Medical transcription samples	Voice transcription testing

8.2 SOP Dataset Construction

No public SOP dataset exists. We will construct one through the following process:

- Partner with 1–2 hospitals to collect 50–100 real SOPs from high-impact departments (Emergency, ICU, Pharmacy, Infection Control, Blood Bank).
- Augment with publicly available clinical guidelines: WHO, CDC, AHA/ACLS, NICE, and Surviving Sepsis Campaign protocols.

- 3. Parse and structure each SOP into a JSON schema containing ordered steps, preconditions, time constraints, numerical thresholds, and contraindication rules.
- 4. Generate 20–50 QA pairs per SOP using LLM-assisted generation, validated by clinical experts. Target: 2,000+ validated QA pairs.
- 5. Create an adversarial evaluation set with deliberately incorrect responses for each violation type (threshold, sequence, contraindication) to benchmark the Procedural Faithfulness Verifier.

8.3 Evaluation Metrics

Dimension	Metric	Method
Retrieval Quality	Recall@5, MRR	Gold passage matching
Answer Correctness	Accuracy, F1	Expert-labeled QA set
Procedural Faithfulness	Violation Detection P/R/F1	Adversarial violation set
Hallucination Rate	Faithfulness Score	RAGAS framework
Clinical Safety	Safety Pass Rate	Expert review (n=200)
Voice Accuracy	WER on Medical Terms	Medical transcription test set
Usability	SUS Score, Task Time	Clinician user study

9. Expected Contributions

- **Procedural Faithfulness Verification Framework:** A novel verification approach that checks LLM-generated clinical responses for threshold accuracy, step sequence correctness, and contraindication completeness — going beyond existing semantic faithfulness metrics.
- **Agentic Clinical RAG Architecture:** A reusable multi-agent design pattern for safety-critical QA systems with built-in verification feedback loops and confidence-gated output.
- **Structured SOP Schema:** A JSON-based representation format for clinical SOPs that enables both AI-powered retrieval and automated procedural verification.
- **Adversarial Procedural Violation Dataset:** A benchmark dataset of clinically incorrect responses annotated with specific violation types, publishable as an open resource.
- **Clinician-in-the-Loop SOP Management:** A practical framework for structured SOP updates with version tracking and automatic knowledge base synchronization.
- **Deployable System:** A complete, open-source system designed for real hospital deployment without EHR integration — reducing the primary barrier to clinical AI adoption.

10. Proposed Timeline

Phase	Duration	Activities
Phase 1	Months 1–2	Literature review, SOP collection, dataset construction, JSON schema design
Phase 2	Months 3–4	Core RAG pipeline, hybrid retrieval, structured SOP parsing
Phase 3	Months 5–6	Procedural Faithfulness Verifier, agentic pipeline (LangGraph), adversarial test set

Phase 4	Month 7	Voice input integration (Whisper), SOP update portal with version tracking
Phase 5	Month 8	Frontend development, feedback loop, confidence scoring, explainability layer
Phase 6	Months 9–10	Evaluation: automated metrics, expert clinical review, usability study
Phase 7	Months 11–12	Thesis writing, revisions, submission

11. Conclusion

This thesis proposes SOP-Guard, an agentic RAG system that addresses a critical gap in clinical AI: the absence of procedural faithfulness verification in LLM-generated responses to clinical protocol queries. By combining hybrid retrieval, multi-agent orchestration, automated verification of thresholds, sequences, and contraindications, voice-based input, and a clinician-driven SOP update mechanism, SOP-Guard is designed to be both a meaningful research contribution and a practically deployable tool.

The system's deliberate avoidance of EHR integration, support for on-premise deployment, and incremental adoption model address the most common barriers to hospital AI deployment. Its verify-then-respond architecture establishes a design principle that we argue should be standard for any AI system operating in safety-critical domains.

We anticipate that the Procedural Faithfulness Verifier will detect 85–95% of procedural errors that standard RAG systems would miss, with total response times under 5 seconds. The structured SOP schema, adversarial evaluation dataset, and verification framework will be published as open resources to advance the field of safe clinical AI.

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